EAB 770 - Chapter 1

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1 Chapter 1: Introduction

1.1 Outline

- Syllabus Discussion
- Accessing SAS OnDemand for Academics
- Why are we here?
- Preliminary Materials

1.2 Syllabus

- The syllabus can be accessed through
  - Canvas
  - https://madfudolig.quarto.pub/eab-770-applied-statistical-methods-for-categorical-data/

1.3 Programming Language

- We will primarily use SAS for this course.
  - You can access SAS through SAS On Demand for Academics. This will be free and accessible when you have internet connection.

1.4 Accessing SAS On Demand for Academics

- Create a SAS Profile that will be used to register for SAS OnDemand for Academics. To register, visit https://www.sas.com and fill out the form to *Create profile*.
  - Use your UNLV email for faster verification.
- Create your account for SAS OnDemand for Academics. To register, visit https://welcome.oda.sas.com sign in with your SAS Profile and complete the steps to *Register for an account*.

2 Why are we here?

Why bother learning categorical data analysis?

3 Review Questions

3.1 Review Questions

Define the following terms:

- p-value
- confidence level in confidence intervals
- Odds
- Relative Risk

3.2 Review Questions

Define the following terms:

- p-value
  - The probability of recording a test statistic as extreme or more extreme as the obtained statistic *assuming the null hypothesis is true.*
- Confidence level in confidence intervals
  - The **long-run proportion** of confidence intervals that theoretically contain the true parameter value.

- Odds
  - The ratio of the success and failure probabilities.
  - $\frac{p_{success}}{p_{failure}}$
- Relative Risk
  - The ratio of success probabilities of two different groups.
  - p_1/p_2

4 Categorical Data

4.1 Categorical Data

Categorical data is present everywhere!

- Demographics
- Opinions
- Ratings
- Locations

4.2 Types of Categorical Response Variables

- Dichotomous
  - Yes/No
- Ordinal
  - More than two possible outcomes with inherent ordering
- Nominal
  - More than two possible outcomes without inherent ordering
- Discrete Counts
  - Numerical outcomes consisting of counting numbers.
- Grouped Survival Times
  - Number of patients who fail/survive during a specific time interval.
  - The interval of failure/survival (time-to-event) is categorical.

4.3 Sampling Frameworks

- Historical data
  - Observational data, usually secondary data
  - No randomization
  - Example: Public data sets.
- Experimental Data
  - Studies that involve random assignment of treatments to different subjects.
  - Random assignment/allocation
  - Example: Randomized Controlled Trials
- Sample Survey Studies
  - Randomly sampled from a larger study population
  - Random selection

4.4 Effect of Randomization

- Randomization affects conclusions of generalization and causation.
- Historical data
  - No randomization: hard to generalize unless the data was randomly sampled from a predetermined population
- Experimental Data
  - Random assignment: balances effects of confounding variables
- Sample Survey Studies
  - Random sampling: representative sample provides good coverage of the larger population

4.5 Randomization

Randomization can be done per subject or a cluster of subject.

- Strata (survey studies) or blocks (experiments)
- Examples
  - Stratified random samples
  - Randomized block designs
  - Multi-level experimental designs

4.6 SAS Procedures and Data

- There are multiple ways to import data into SAS
  - DATA step
  - PROC IMPORT step
  - SAS OnDemand can import data from CSV files.
- `FREQ`
  - Contingency tables (2×2, $s \times r$)
  - Chi-squared tests
- `LOGISTIC`
  - Logistic Regression
- `GENMOD`
  - Generalized Estimating Equations
- Other procedures include `GLM` and `CATMOD`.

4.7 SAS in Class

- Sample code will be provided, but there will be examples where we work the code out together as a class.
- Class computers have SAS installed in them.
  - You can save files on your Y drive (assigned to your ACE ID) to access files on other computers.
  - SAS can also be accessed through SAS OnDemand Academics.
    - Needs internet connection

4.8 SAS Examples

- Download Chapter1.sas from Canvas.

4.9 SAS Example

```
data respire;
length treat $7. outcome $1. ;
input treat $ outcome $ count;
datalines;
placebo f 16
```

```
placebo u 48
test f 40
test u 20
;
```

- ALL lines should be ended with a semi-colon **;**
- `data` statement is followed by the data set name.
- `length` specifies the number of characters included in each **character**.
- `input` names all variables included in the data set
  - The `$` after the variable tells SAS that these variables are categorical variables.
- `datalines;` tells SAS that you are about to add new data points.
  - Each row represents a data point in this data table.
  - End with a separate **;**

4.10 SAS Example: Contingency Table

```
proc freq data=respire;
weight count;
tables treat*outcome;
run;
```

- `proc freq` calls the `FREQ` procedure in SAS.
  - Not defining `data=respire` means SAS will automatically use the most recently run data set.
- `weight` tells SAS the number of counts for each count
- `tables` creates the contingency table with row, column, and total percentages.
- `run;` tells SAS to run the code. **This is important!**

4.11 SAS Example: One Observation per Row

```
data severe;
input treat $ outcome $ @@;
datalines;
placebo f placebo f placebo f
placebo f placebo f
placebo u placebo u placebo u
placebo u placebo u placebo u
placebo u placebo u placebo u
placebo u
```

```
test f test f test f
test f test f test f
test f test f
test u test u test u
test u test u test u
test u test u test u
test u test u test u
test u test u test u
test u test u test u
test u test u
;
```

- This is an example of a data set where one row corresponds to one observational unit.
- The `@@` sign means that the data lines contain multiple observations. `@@` basically means new line after two entries written in the `datalines` statement.

4.12 SAS Example: Contingency Table

```
proc freq data=severe;
tables treat*outcome / nocol nopct;
run;
```

- The `weight` statement is not needed in the `FREQ` procedure when the data does not consist of aggregate counts.
- `nocol` suppresses the column percentages in the output
- `nopct` suppresses the total percentages in the output
- `norow` suppresses the row percentages in the output

5 Exercise

5.1 SAS Exercise

Suppose a survey asking sophomores and senior students whether they agree with the statement "I enjoyed my in-person classes in AY 2023-2024." yielded the following results.

Enter the data in SAS using the data step and generate the contingency tables. Name the data set `enjoy`.

	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
Sophomore	24	43	12	25	12
Seniors	19	37	24	32	21

5.2 SAS Exercise

There are many ways to code the `data` step. Here's one way to do it!

```
data enjoy;
length year $9. response $2.;
input year $ response $ count @@;
datalines;
sophomore SA 24 sophomore A 43
sophomore N 12
sophomore D 25 sophomore SD 12
senior SA 19 senior A 37
senior N 24
senior D 32 senior SD 21
;

proc freq order=data data=enjoy;
weight count;
tables year*response;
run;
```